

Shreya Pramanik

Minneapolis, MN

📞 952-245-5379

✉ prama018@umn.edu

in LinkedIn

🐙 Github

🌐 Portfolio Website

Education

University of Minnesota

Master of Statistics

Coursework: Data Mining, Computer Vision, Regression Analysis, Advanced Probability, Bayesian Analysis

Sep 2023 – Aug 2026

Minneapolis, United States

Indian Institute of Technology Kanpur

Master of Statistics

Coursework: Statistical Simulation, Time Series Analysis, Design of Experiment, Applied Stochastic Process

Sep 2020 – Jun 2022

Kanpur, India

Presidency University

Bachelor of Statistics

July 2017 – July 2020

Kolkata, India

Technical Skills

Languages & Frameworks: Python, R, SQL, NOSQL, SAS, MATLAB, Excel, Power BI (DAX, Power Query), Tableau, PyTorch, TensorFlow, scikit-learn

AI/ML & NLP: LangChain, LangSmith, RAG, ChromaDB, Hugging Face, Prompt Engineering, Diffusion Models

Cloud & DevOps Tools: Docker, Git, Streamlit, AWS, Azure, Databricks, Airflow, n8n, Claude Code, ETL Pipelines

Experience

HSBC Electronic Data Processing Pvt. Ltd

Data Analyst, Wealth and Personal Banking

July 2022 – July 2023

Bengaluru, India

- Queried and extracted large-scale customer contact datasets via **SQL** across 5+ APAC contact centers; built Excel-based reporting tools using **pivot tables**, **lookup formulas**, and automated PowerBI **KPI summaries** to clean, structure, and surface trends.
- Developed and deployed an ensemble of short-term demand forecasting models using **XGBoost**, **LSTM**, and **Prophet** across 5+ APAC contact centers, achieving **80–86%** accuracy, informing real-time staffing decisions for hundreds of daily customer interactions.
- Led production-ready modeling pipeline on Malaysia customer-contact data to detect demand surges during digital-banking rollouts, informing **workforce planning decisions** to business stakeholders.

School of Statistics, University of Minnesota

Graduate Teaching Assistant

Sep 2023 – Jan 2026

Minneapolis, MN

- Led coding labs for regression and simulation courses; mentored 100+ grad students on statistical programming in R.

Carlson School of Management, University of Minnesota

Graduate Research Assistant

Feb 2026 – May 2026

Minneapolis, MN

- Cleaned 654 survey responses (flagged ~5% low-quality) and coded 16 interview transcripts (**Cohen's $\kappa = 0.78$**) for a longitudinal reduced work-week policy study.

Projects

Telecom Customer Care RAG Assistant | *LangChain, ChromaDB, LangSmith, Groq, Streamlit* [🔗](#) [🐙](#) Aug 2026

- Deployed an end-to-end **customer-facing RAG chatbot** to Streamlit Cloud, serving telecom support queries in **under 4 seconds**, by architecting a **LangChain pipeline** that ingests 3 heterogeneous sources (CSV FAQ, SQLite tickets, PDF guide) into **ChromaDB collections** and generates grounded responses through **Qwen 3.6 27B on Groq**.
- Eliminated PII exposure and off-topic responses by implementing **dual guardrail layers** (input: regex PII masking, topic gating; output: disclaimer injection, hallucination flagging), wrapped as **RunnableLambda** components in the LCEL chain and fully traced in **LangSmith**.
- Achieved **0.95 faithfulness** and **1.00 conciseness** by building a repeatable **LangSmith experiment pipeline** with 3 custom scorers benchmarked against human-written resolutions extracted from resolved support tickets, enabling measurable prompt and retriever iteration across experiment runs.

E-Commerce Analytics Platform | *MySQL, Python, Power BI, DAX* [🔗](#) [🐙](#)

Jun 2026 – Jul 2026

- Built an end-to-end **ETL pipeline** using Python (pandas, SQLAlchemy) to ingest **100K+ real transactions** across a 3-layer medallion architecture (staging → 3NF → star schema), normalising raw Olist e-commerce CSVs into **8 relational tables** with FK constraints enforced at the database level.
- Designed a **5-page Power BI dashboard** tracking **\$13.17M in revenue** across 95K orders, identifying checkout as the

highest drop-off stage (**11.56%**), organic search as the top-performing channel (**62x more efficient** than paid search), and VIP customers spending **2.3x more** than first-time buyers; supported by 6 production-grade SQL queries using window functions, recursive CTEs, and NTILE-based RFM segmentation.

- Engineered an **A/B testing framework** with deterministic variant assignment via MD5 hashing in Python and validated a statistically significant **31% conversion lift** (11.71% $\rightarrow$ 15.36%, $z = 9.24 > 1.96$) through a **SQL-based z-test**, while implementing **CI/CD via GitHub Actions** with automated SQL linting and pytest on every push.

LLM-Powered Blood Work Report Analysis | *LangChain, Google Gemini, Streamlit, PyPDF* [🔗](#) [🔊](#) Aug 2026

- Deployed a **two-stage LangChain pipeline** on Streamlit Cloud that extracts every blood test value, classifies it against reference ranges, and generates clinician-style health summaries with dietary guidance via **Google Gemini (Gemma 3 27B)** in **under 10 seconds**.
- Built a unified ingestion layer supporting **PDF and free-text** inputs using **PyPDF**, converting unstructured clinical reports into **structured tabular output** (test name, result, unit, status, reference range) without manual parsing.
- Engineered a **domain-specific prompt chain** that maps flagged blood markers to **specific, affordable Indian foods** (e.g., high LDL $\rightarrow$ oats, methi; low HDL $\rightarrow$ walnuts, flaxseed), delivering **culturally relevant diet plans** personalized to each report.

Prompt-Engineered Expense Automation Pipeline | *n8n, Gemini API, JavaScript* [🔗](#) [🔊](#) Mar 2026 – Apr 2026

- Designed and iterated on **structured prompts for Google Gemini** to classify raw bank transaction descriptions into **7 spending categories**, replacing a fully manual expense-review process and reducing per-batch classification time from **hours to under 5 minutes**.
- Built an end-to-end automation pipeline in **n8n** connecting Google Drive, Gemini API, Google Sheets, and Telegram, with **JavaScript-based** normalization logic to standardize merchant names, generate consistent keys, and deduplicate entries across transaction text variations.
- Implemented a **Telegram-callback-based** review workflow with recursive queue logic for one-at-a-time merchant verification across 7 categories, and integrated a downstream finance dashboard in **Lovable** for real-time expenditure visualization.

Financial Shock Effects on Hospital Operations | *Python, XGBoost, SHAP* [🔗](#) [🔊](#) Jan 2026 – Feb 2026

- Engineered **15+ predictive features** from **66,679** HCRIS hospital-year cost reports (Medicaid share, uncompensated-care burden, occupancy, staffing intensity) and trained an **XGBoost** model with hospital-level grouped cross-validation to forecast operating margins across **6,483 hospitals**, achieving **0.084 MAE** and **0.129 RMSE** on held-out hospitals with **SHAP** surfacing occupancy and Medicaid exposure as dominant drivers.
- Stress-tested every hospital under simulated **5–15% Medicaid reimbursement shocks**, quantifying that low-exposure private urban hospitals (Medicaid <20%) absorbed shocks within **1.2 percentage points** of margin while high-exposure urban teaching hospitals (Medicaid 40–60%) turned financially fragile.
- Scored and ranked all hospitals by a composite resilience metric (Δ margin, post vs. pre-shock), isolating the **top and bottom 20** institutions to give policymakers a data-driven shortlist of the most and least shock-absorptive hospital profiles across **2011–2022**.

Masked Conditioning for Cascaded Diffusion | *PyTorch, U-Net, Diffusion Models* [🔗](#) [🔊](#) Sep 2025 – Dec 2025

- Proposed and implemented a channel-masked conditioning variant of the **cascaded diffusion framework** ($64 \times 64 \rightarrow 128 \times 128$), selectively corrupting **RGB** channels with mask probability 0.35 to improve robustness in **super-resolution** image generation.
- Engineered a two-stage class-conditional diffusion pipeline with **U-Net backbones**, cosine noise schedule over $T=1000$ timesteps, EMA model averaging (decay=0.995), and **DDIM sampling** (200 steps, $\eta=0$), achieving $FID \sim 42$ at 100 inference steps on a 30-class ImageNet subset ($\sim 33K$ images).
- Concluded both baseline and masked conditioning converge to comparable fidelity, suggesting the channel-masking structure preserves conditioning quality without degrading the augmentation benefit.

Harmful Brain Activity Classification | *EfficientNet, TensorFlow, scikit-learn* [🔗](#) [🔊](#) Feb 2025 – May 2025

- Reduced misclassification of 6 harmful brain activity types by **39%** (KL divergence $0.85 \rightarrow 0.52$) by engineering a **dual-source 8-channel spectrogram pipeline** that fused Kaggle-provided spectrograms with EEG-derived mel spectrograms across **17,089 patient recordings** from the HMS Kaggle competition.
- Accelerated neurologist-grade EEG labeling across **4 brain regions** (LL, RL, LP, RP) by transforming **19 raw electrode-pair signals** into clinically interpretable spectrograms using librosa mel-spectrograms with bipolar montage averaging, log-scaling, and z-score normalization at 200 Hz sampling rate.
- Achieved a best cross-validated KL divergence of **0.52** across **5-fold GroupKFold** (grouped by patient ID to prevent leakage) by training an **EfficientNetB0** classifier on combined 512×512 spectrogram images with cosine LR annealing and mixed-precision inference on dual T4 GPUs.

Auto Insurance Claim Prediction | *R, XGBoost, Tweedie Regression, Cross-Validation* Oct 2024 – Dec 2024

- Led **R-based modeling and feature selection** for a 60,000-record insurance claim-cost prediction project, handling a highly **semi-continuous dataset** with only **6.8% non-zero claims** and positive costs from \$200 to $\sim$ \$60K.

- Compared Logit + Gamma, Tweedie regression, and XGBoost models using an 80/20 train-test split, 5-fold **cross-validation**, over-dispersion checks, Hosmer-Lemeshow testing, and **AIC-based variable selection**.
- Improved final normalized **Gini index to 0.45** and presented **stakeholder-ready risk drivers**, including exposure, vehicle value, max power, term length, vehicle age, and engineered ratio features for **claim-cost planning**.

Large-Scale Twitter Discourse Analysis: Russia-Ukraine Conflict | NLP, K-Means [🔗](#) [📢](#) Jan 2022 – Apr 2022

- Engineered an end-to-end Twitter NLP pipeline that scraped **2.9M tweets** over **15 days** and filtered to **943K English posts**, cutting downstream processing noise by **~67%**.
- Reduced high-dimensional text feature space across **5 NLP representations** (BoW, n-gram, TF-IDF variants) using **PCA**, retaining **90% variance** while shrinking feature complexity for K-Means clustering.
- Identified **3 dominant discourse themes** in ~943K war-related tweets using **silhouette- and elbow-optimized K-Means**, surfacing public sentiment patterns across geopolitical, humanitarian, and nuclear threat narratives.

Achievements

- **Won HeatMap Hackathon 2026** (BData Inc. × American Burn Association × MN Healthcare), built a geospatial systems solution translating burn injury data into an equitable care access analysis, selected over competing teams for real-world strategic impact.
- Received **Academic Excellence Award** for outstanding performance in MSc Statistics, IIT Kanpur.